

Synthetic Control Method

Application: Sovereign Wealth Funds and Economic Growth



UNIVERSITY OF
SOUTH ALABAMA

Background

Sovereign Wealth Funds

- State-owned investment fund usually derived from a country's surplus reserves
- Norway Government Pension Fund Global, China Investment Corporation, Hong Kong Monetary Authority Investment Portfolio, Kuwait Investment Authority,..., 89 current SWF worldwide.
- Total Assets ~ \$8.2T (source: <https://www.swfinstitute.org/> - 2020)
- Economy Theory suggests that a SWF should be beneficial
- Empirical evidence shows mixed results (Bagattinti, 2011)

Sovereign Wealth Funds

Macroeconomic Objectives

- Objective #1: to create a capital base for the growth and preservation of national wealth by investing excess reserves into diversified assets
- Objective #2: to stabilize the macroeconomic environment by using the SWF to decrease the volatility of government spending during crisis periods
- Objective #3: to promote economic development by investing in capital assets (energy, transport, water management, communication, etc.) and social infrastructure (e.g. education, health, human capital)

Development Economics Research Motivation

UN Millennium Development Goals

- Goal #1: End Poverty
- Goal #2 : Eradicate Hunger
- Goal #3 : Good Health and Well-Being
- Goal #6: Clean Water and Sanitation
- Goal #7: Affordable and Clean Energy
- Goal #9: Industry, Innovation, and Infrastructure

Development Economics Research Motivation (Cont'd)

Small Island Development States (SIDS)

- Focus of the UN Millenium Goals (2005)
- Samoa Pathway (Apia, Samoa, 1-4 September 2014): Third International Conference on SIDS: “**Sustainable development for the SIDS**”
- Caribbean Countries (SIDS) characteristics:
- Blue Economies: Sustainable Use of Ocean Resources
- Crossroads: North and South America
- Transition from agriculture or mining to a service-driven economy
- Extremely vulnerable to climate change and natural disasters

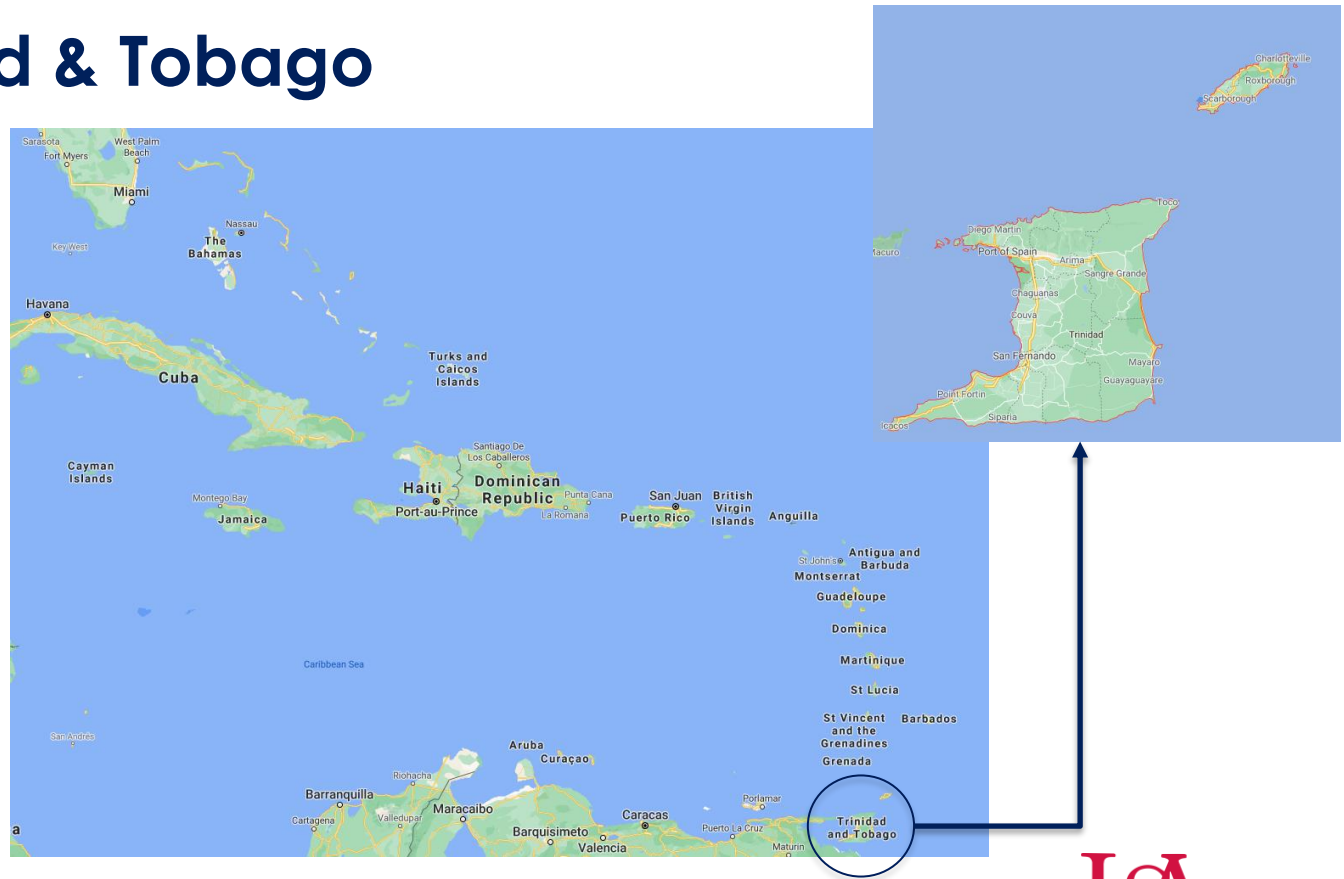
Research Question

Does the existence of a SWF can influence the economic growth of its host economy?

(Unique Contribution)

Case Study: SIDS

Trinidad & Tobago



Case Study: Trinidad & Tobago

Trinidad & Tobago

- The **Interim Revenue Stabilisation Fund** (IRSF), the SWF of T&T was first announced in 1997 (US\$415 million).
- IRSF was created as a “rainy day” fund in T&T on September 29, 2000
- the **Heritage and Stabilisation Fund** (HSF) was founded in March 2007 with a balance of US\$1.4 billion which was transferred from the IRSF.

HSF objectives:

Objective #1: a stabilization component to protect fiscal policy from large fluctuations in energy sector revenues

Objective #2: savings component to benefit of future generations

Heritage and Stabilisation Fund

Insights

- The HSF is invested in medium to long-term foreign currency denominated assets, fixed income securities and equities
- The board takes decision about the strategic asset allocations. The fund is operated with a view to achieving a target rate of return.
- 60% of the difference between the actual tax revenue and the estimated tax revenue from oil and gas are accumulated to the HSF annually.
- Total Assets: US\$6.48 billion (December 31, 2019) - 82.10% invested in the US financial market
- The portfolio quarterly return was 1.05% in the last quarter of 2019

Heritage and Stabilisation Fund

Covid 19

- T&T has decided to use part of the fund to facilitate economic recovery during the post-COVID period through different programs such as:
- salary relief grants
- food cards
- trade payables
- VAT refunds
- income tax refunds

Data (source: The World Bank)

Table 1. Descriptive Statistics (Between Economies)

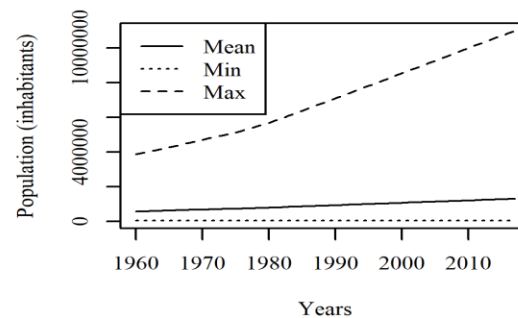
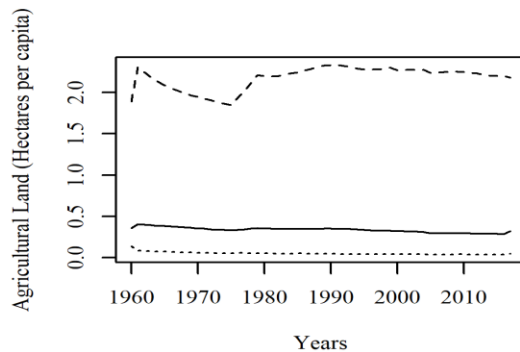
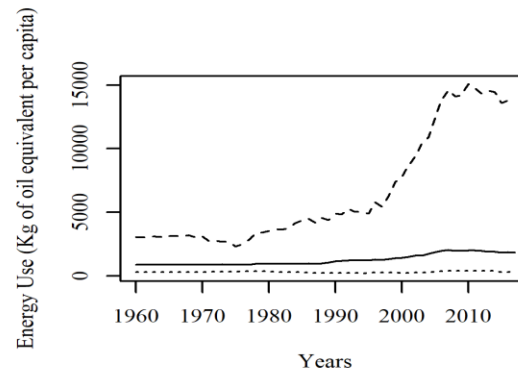
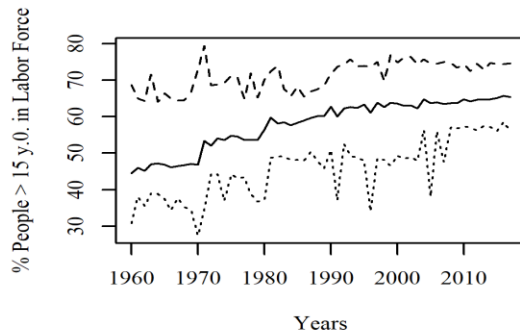
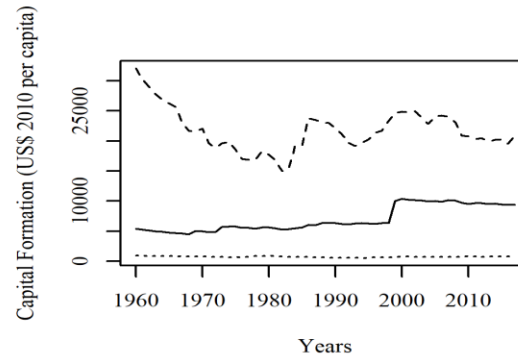
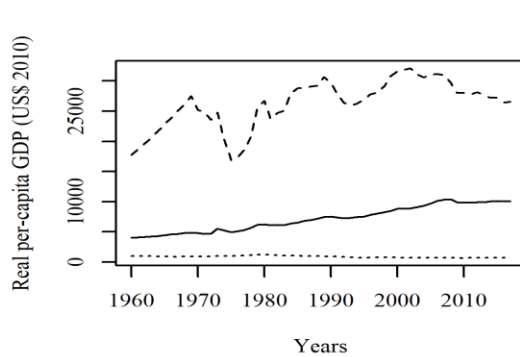
Economy	GDP	Capital	Labor	Energy	Land	Population
Antigua and Barbuda	8,475.7 (3,875.5)	7,842.8 (3,307.1)	59.5 (10.48)	964.6 (516.7)	0.12 (0.03)	76,218 (12,822)
The Bahamas	26,251 (3,963.7)	21,767 (3,495.6)	67.8 (7.87)	2,172.4 (56.65)	0.05 (0.02)	252,031 (81,674)
Barbados	2,767.7 (1,231.9)	2,929.6 (804.4)	52.07 (9.14)	580.3 (73.82)	0.62 (0.12)	20,1247 (84,034)
Belize	11,162.9 (5,739)	10,574.8 (5,080.3)	60.83 (10.46)	988.4 (333.65)	0.07 (0.02)	258,646 (16,466)
Dominica	4,637.9 (1,369.4)	7,708.7 (4,917.6)	52.63 (8.59)	516.7 (36.33)	0.28 (0.04)	70,765 (3,383)
Grenada	4,772.1 (2,077.4)	5,438.7 (3,246.2)	57.77 (11.38)	564.7 (87.15)	0.15 (0.06)	98,070 (5,361)
Guyana	2,210.1 (633.9)	1,743 (1,355.9)	54.21 (6.11)	631.3 (19.37)	2.15 (0.3)	735,187 (49,104)
Haiti	884.5 (158.3)	733.2 (104.2)	62.48 (4.49)	300.4 (59.73)	0.26 (0.08)	7,070,037 (2,198,195)
Jamaica	4,443.9 (473.8)	3,693.3 (1,092.1)	64.57 (4.34)	1,148.4 (211.18)	0.22 (0.06)	2,344,862 (397,320)
Saint Kitts and Nevis	9,515.3 (4404.9)	11,480.3 (6,687.1)	56.81 (11.15)	792.2 (475.25)	0.27 (0.1)	46,371 (4,153)
Saint Lucia	5,299.1 (2326.9)	4,518.7 (1,643.1)	59.74 (10.58)	597.9 (84.36)	0.13 (0.05)	134,895 (28,566)
Saint Vincent and The Grenadines	3,703.8 (1,818.4)	5,265.4 (3,085.1)	55.66 (11.66)	546.3 (73.07)	0.11 (0.02)	101,484 (9,077)
Suriname	6,277.2 (1,419.1)	7,225.2 (2,523.2)	50.67 (7.15)	1,006.8 (264.49)	0.18 (0.05)	420,697 (76,827)
Trinidad and Tobago	9,714.1 (3,876.2)	6,533.5 (2,487.6)	56.18 (7.05)	6,641.8 (4,462.16)	0.10 (0.24)	1,152,857 (162,178)

**Panel of 812 observations
from 14 economies (the
founding members of
CARICOM) from 1960 to
2017**

Notes: GDP: real per capita GDP (expressed in constant 2010 US\$); Capital: per capita capital formation (expressed in constant 2010 US\$); Labor: percentage of people older than 15 years in the labor force; Energy: energy use (Kg of oil equivalent per capita); Land: per capita agricultural land (hectares/person); Population: number of inhabitants. Means and (Standard Deviations) over 1960 – 2017. Sample size = 812 observations.



Data (source: The World Bank)



**Descriptive Statistics
(within CARICOM Economies)**



Research Method

Synthetic Control Method for Comparative Case Study

[Abadie, Diamond & Hainmueller, 2010]

- Allows observational (nonrandomized) study to mimic the characteristics of a randomized controlled trial
- Causal Inference
- Nonparametric generalization of a Difference-In-Difference (DID) estimator

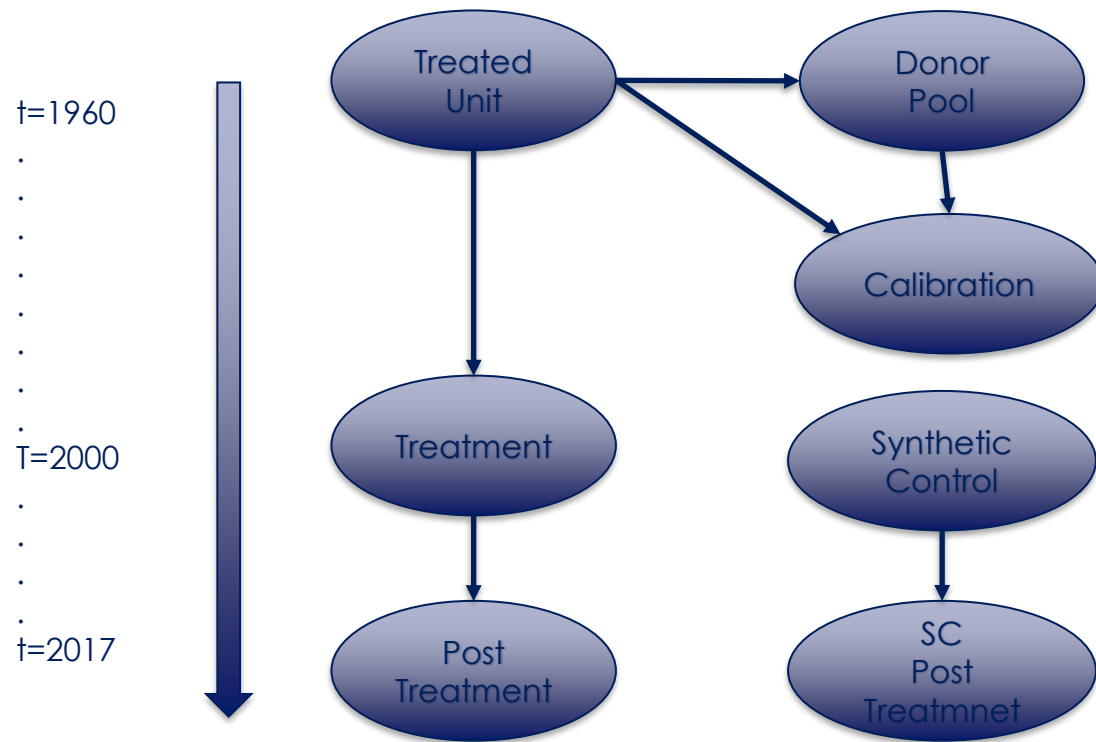
Synthetic Control Method

In a nutshell

- Trinidad and Tobago = Treated Unit
- Remaining 13 CARICOM economies (Donor Pool) = Control Units
- Treatment = SWF
- The treated unit received the treatment in 2000
- Find a convex combination of control CARICOM economies that resembles the economy of Trinidad and Tobago.
- Analyze the difference between Trinidad and Tobago and Synthetic Counterfactual Trinidad and Tobago after treatment

Synthetic Control Method

Concept Map



Synthetic Control Method

Rigorous Formulation

- observed the pre-intervention periods $t = 1960, 1961, \dots, 2000$ and the post-intervention periods $\tau = 2001, 2002, \dots, 2017$.
- two possible GDP outcomes for unit i in year t : Y^0_{it} , i.e., GDP of unit i in year t if the economy was not exposed to the intervention, and Y^1_{it} , i.e., GDP of unit i in year t if the economy was exposed to the intervention.
- Impact of SWF (Post Intervention): $\delta_\tau = Y^1_\tau - Y^0_\tau$ [difference-in-differences panel estimator with fixed-effects varying over time]
- Note: We Do Not Observe Y^0_τ , we will recover the unobserved outcome as the convex combination of the observed outcomes from the donor pool:

$$\hat{\delta}_\tau = Y^1_\tau - \sum_{j=1}^{13} \hat{w}_j Y^0_{j\tau},$$

Synthetic Control Method (Cont'd)

Rigorous Formulation

$$\operatorname{argmin}_{\hat{w}_j \in W} \sqrt{(\Gamma^1 - \Gamma^0 W)' \Lambda (\Gamma^1 - \Gamma^0 W)} \quad \text{with } w_j \geq 0 \text{ and } \sum_j w_j = 1$$

$\Gamma^1 = f[(\text{Capital/POP}), \text{Labor}, \text{Energy}, (\text{Land/POP})]$ special predictors for treated unit

$\Gamma^0 = f[(\text{Capital/POP}), \text{Labor}, \text{Energy}, (\text{Land/POP})]$ special predictors for control pool

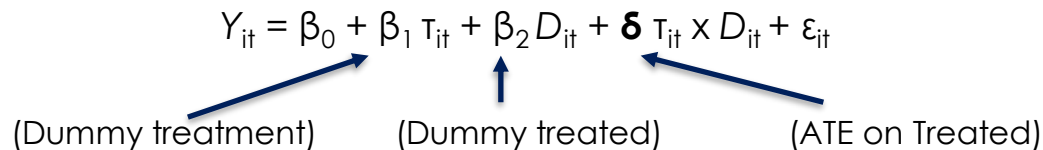
Λ = positive semidefinite matrix which allows weights to be function of the variables predicting power

Difference-in-Difference

Sensitivity Analysis

$$Y_{it} = \beta_0 + \beta_1 T_{it} + \beta_2 D_{it} + \delta T_{it} \times D_{it} + \varepsilon_{it}$$

(Dummy treatment) (Dummy treated) (ATE on Treated)



In comparative studies it is a good practice to allow the starting time of the intervention to be in the neighborhood of $\{\tau-5, \tau-4, \dots, \tau-1, \tau, \tau+1, \dots, \tau+4, \tau+5\}$, i.e., 5 years before and after the intervention

Difference-in-Difference Results

Table 2. Difference-in-differences Models for different intervention starting periods.

τ	δ	AIC	BIC	RMSE	σ_ε
1995	2,578 (1,789)	16,549	16,573	6,407	6,411
1996	2,877 (1,803)	16,549	16,572	6,406	6,410
1997 ^a	3,176* (1,821)	16,549	16,572	6,405	6,409
1998	3,469* (1,841)	16,549	16,572	6,407	6,411
1999	3,753** (1,866)	16,550	16,573	6,409	6,413
2000 ^b	4,027** (1,864)	16,551	16,574	6,414	6,418
2001	4,303** (1,927)	16,553	16,576	6,422	6,426
2002	4,577** (1,965)	16,554	16,578	6,429	6,411
2003	4,826** (2,008)	16,556	16,580	6,436	6,409
2004	4,980** (2,058)	16,559	16,582	6,447	6,410
2005	5,091** (2,116)	16,562	16,585	6,459	6,411

Notes: Standard errors in parentheses; *90%, **95% confidence; $\tau(0/1)$ is a dummy variable that picks value 1 on and after each intervention starting time and zero otherwise; δ = change in annual Trinidad and Tobago per-capita GDP due to intervention; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion; RMSE = Root Mean Square Error; σ_ε = residuals standard deviation (808 degrees of freedom); ^ayear with simultaneous lowest AIC, BIC, RMSE, and σ_ε ; ^byear when the Heritage fund was started.



Synthetic Control Method

Calibration

Table 3. Optimal weights for the Synthetic Trinidad and Tobago.

Economy	Weights (w_i)
Antigua and Barbuda	0.000
The Bahamas	0.102
Barbados	0.000
Belize	0.000
Dominica	0.000
Grenada	0.000
Guyana	0.001
Haiti	0.000
Jamaica	0.278
Saint Kitts and Nevis	0.000
Saint Lucia	0.000
Saint Vincent and The Grenadines	0.000
Suriname	0.619

Note: $w_i = 0$ suggests that the information from the economy data did not contribute to constructing the synthetic control economy; $w_i \neq 0$ suggests that the economy shares similar economic characteristics with the treated unit, thereby working as an information donor to construct the synthetic control economy.

Synthetic Control Method (Cont'd)

Table 4. Real per capita GDP (US\$ 2010) Predictor Means.

Variables	Trinidad and Tobago		Donor Pool (mean)
	Observed	Synthetic	
Real per capita GDP* (1960-1997)	7,316.48	7,214.48	5,752.55
Capital* (1960-1997)	4,893.65	7,662.38	5,590.32
Labor* (1960-1997)	53.40	53.81	54.80
Energy* (1960-1997)	3,712.80	1,081.16	755.80
Land* (1960-1997)	0.13	0.19	0.37

Notes: *special predictors $\subset \Gamma_1$ and Γ_0 obtained as a linear combinations of pre-intervention outcomes which includes per capita GDP and the predictors Capital, Labor, Energy and Land to allow the synthetic control outcome to fall outside of the convex hull of the control units if desired.

Synthetic Control Method (Results)

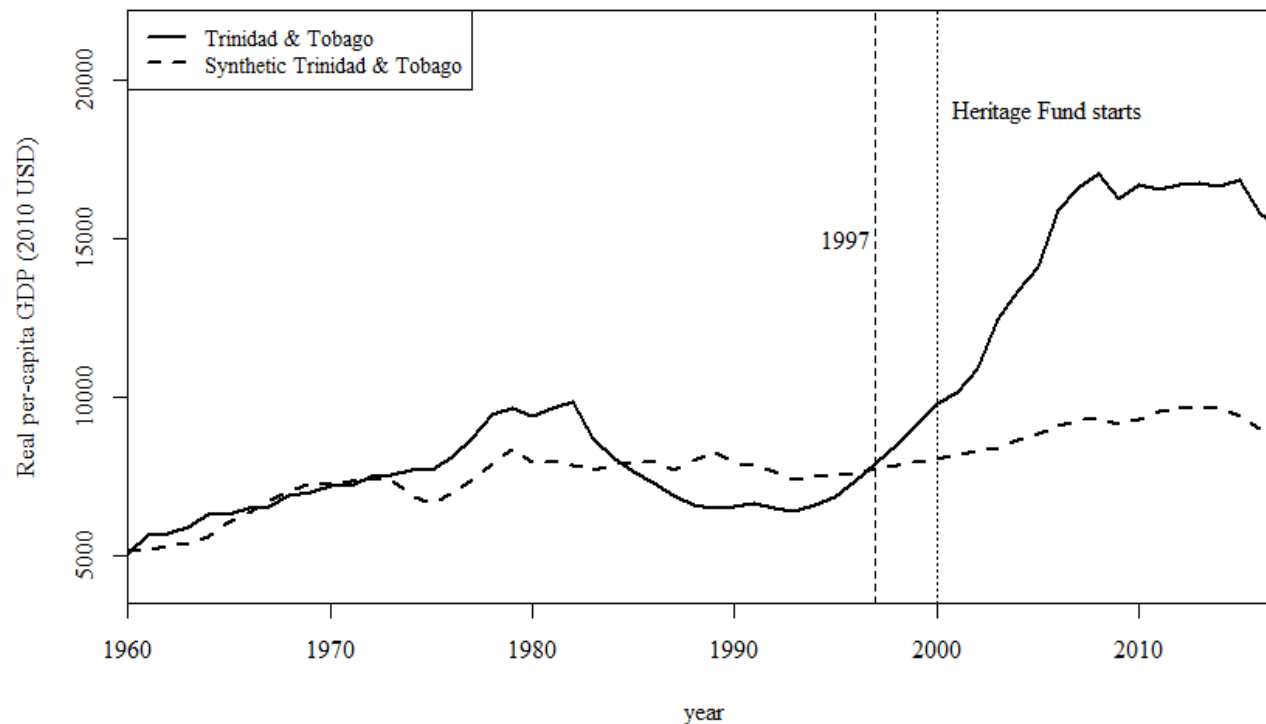
Table 5. Gap in per capita real GDP (US\$ 2010) of Trinidad and Tobago due to the Heritage & Stabilisation Fund.

Year	Actual Trinidad and Tobago	Synthetic Trinidad and Tobago	Gap
1997	7,877	7,752	125
1998	8,505	7,850	655
1999	9,171	7,934	1,237
2000	9,779	8,050	1,729
2001	10,152	8,192	1,960
2002	10,911	8,293	2,618
2003	12,426	8,414	4,012
2004	13,346	8,671	4,675
2005	14,105	8,857	5,248
2006	15,892	9,107	6,785
2007	16,570	9,284	7,286
2008	17,052	9,279	7,773
2009	16,226	9,129	7,097
2010	16,683	9,300	7,383
2011	16,551	9,535	7,016
2012	16,680	9,654	7,026
2013	16,761	9,708	7,053
2014	16,641	9,632	7,009
2015	16,824	9,394	7,430
2016	15,765	8,989	6,776
2017	15,350	9,047	6,303

$\Sigma = \$107,196$

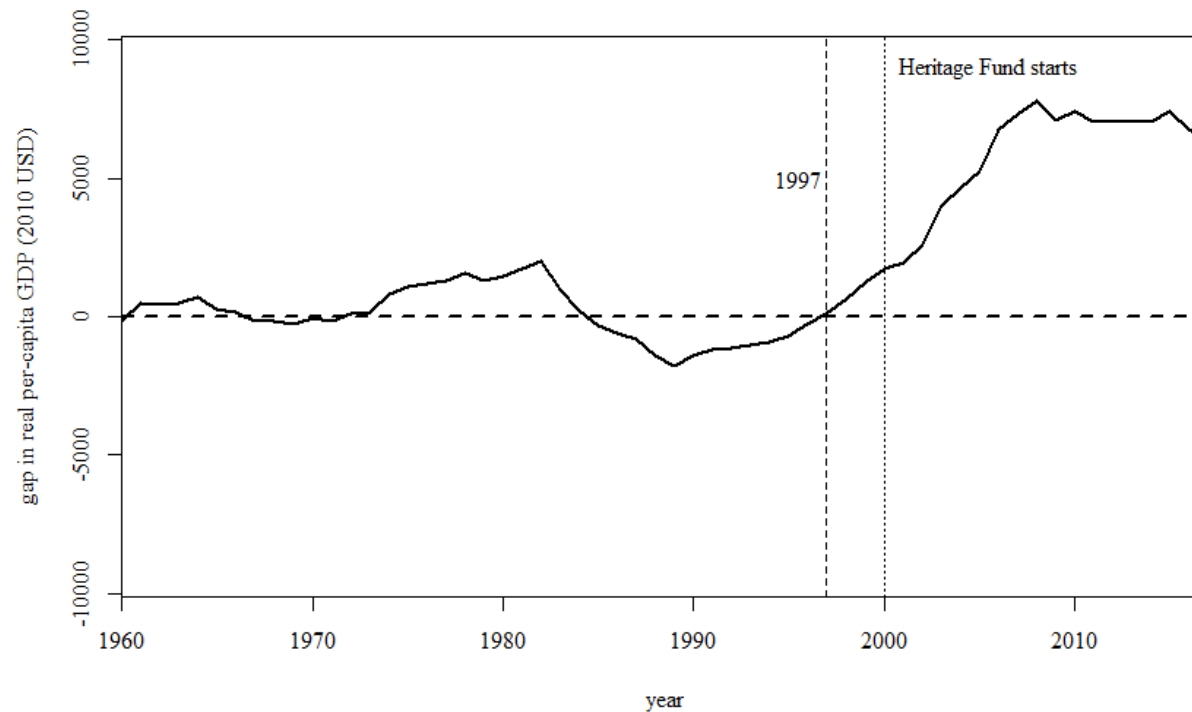
Synthetic Control Method (Results)

Trends in real per-capita GDP (Trinidad and Tobago vs. Synthetic Trinidad and Tobago).



Synthetic Control Method (Results)

Real per-capita GDP gap between Trinidad and Tobago and Synthetic Trinidad and Tobago



Gap =
TT – Synthetic TT

Placebo (or Falsification) Test

Angrist and Krueger (1999); Abadie and Gardeazabal (2003)

- random permutation of the intervention assignment to any unit of the treated and control groups.
- i.e., applying the synthetic control method to any CARICOM economy of the donor pool
- Results: 14 synthetic control CARICOM economies
- Compute the post- to pre-intervention mean squared prediction error (MSPE) ratio

Placebo (or Falsification) Test

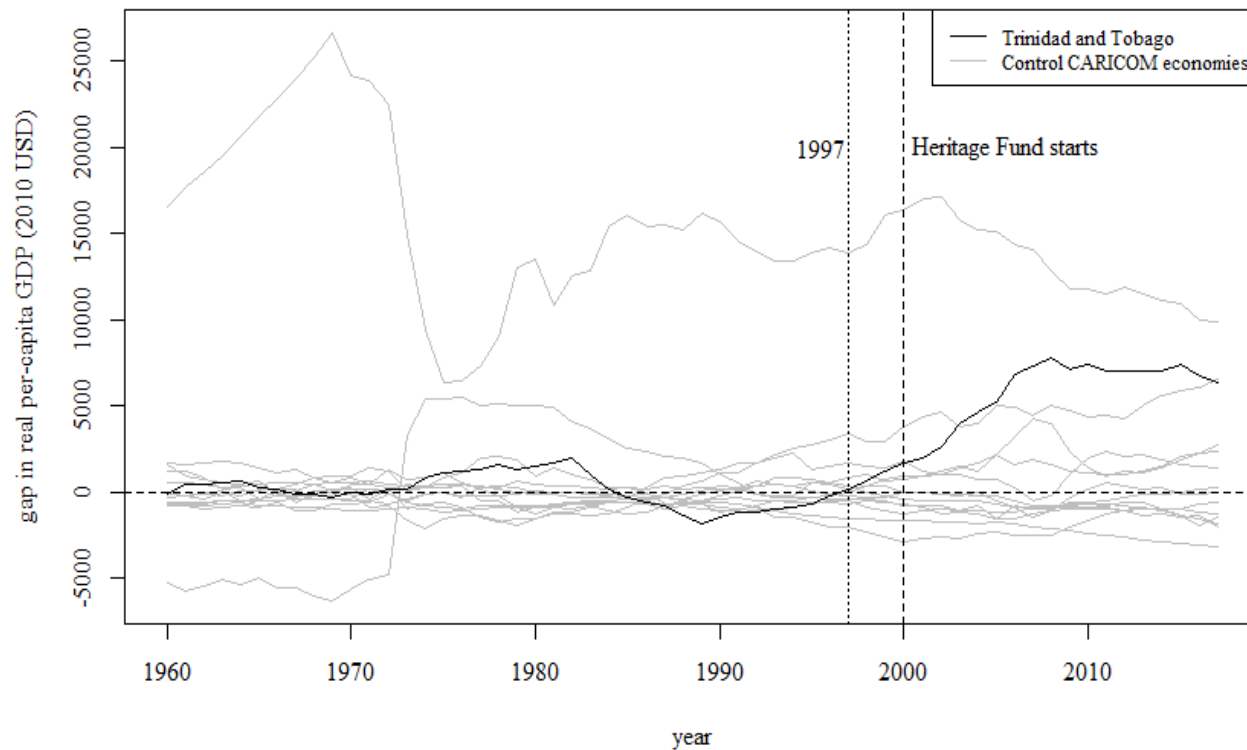


Table 6. Ratio of post/pre-Heritage Fund mean squared prediction error: Trinidad and Tobago and control CARICOM economies.

Economy	MSPE ratio
Antigua and Barbuda	3.039*** (0.846)
Bahamas	0.655*** (0.084)
Barbados	0.096*** (0.023)
Belize	3.54*** (0.982)
Dominica	1.88*** (0.515)
Grenada	30.444*** (7.744)
Guyana	1.66*** (0.455)
Haiti	6.568*** (0.959)
Jamaica	0.912*** (0.199)
Saint Kitts and Nevis	14.662*** (4.177)
Saint Vincent and The Grenadines	1.883*** (0.33)
St. Lucia	2.767* (1.448)
Suriname	3.761*** (0.825)
Trinidad and Tobago	35.656*** (9.505)

Notes:*99%, *90% confidence; bootstrap standard errors in parentheses.**

Placebo Test



Conclusions

- The synthetic control method indicates the average increase in real per capita GDP, between 1997 and 2017, is \$5,104.57 (expressed in constant 2010 US\$) vs. \$3,176.00 (2010 US\$)
- The cumulative welfare change due to the existence of the SWF is estimated at \$107,196.00
- We suggest the government of T&T follow flexible policies to get the greatest benefit from the SWF during the next few years
- According to Statista , the government expenditure was over 34% of the GDP of T&T in 2019. T&T should be careful of large budget deficit that could affect the effectiveness of SWF.

Conclusions (Cont'd)

- We suggest eliminating less productive projects from future budget plans to keep the budget deficit low so that the SWF can meet the deficit and help maintain growth for the economy of T&T
- We suggest small islands, non-renewable resource-dependent, and non-diversified economies should consider the implementation of SWFs.

Thank You



Q&A

Ermanno Affuso, Ph.D.

Department of Economics, Finance, & Finance
314 Mitchell College of Business
University of South Alabama
Mobile, AL 36688 U.S.
Tel. +1 (251) 460 7908
eaffuso@southalabama.edu

